The Task

Item Formation

Your task is to group together **identical products** (called *items*) across or within retailers. Products are considered identical if they share the same GTIN/EAN/Barcode code. Products with the same GTIN are expected to have consistent attributes, such as: *quantity*, *flavour*, *pack type*, *volume*, *container pack type, caffeine content*, *brand*, etc.

The Dataset

The dataset contains **thousands of products (70K+)**, each identified by a unique **SKU_ID**. Each row represents a distinct product scraped from an online retailer. Some product details have been **automatically extracted** using machine learning models, including *Brand*, *Category* and *Attributes* (e.g flavor, caffeine content, etc.).

Please note that these fields **are not perfect** - there are some precision errors, and recall is incomplete. For example, certain attributes may be missing simply because the retailer did not provide that information. As a result, the attribute coverage is **sparse**. Additionally, some fields originate directly from online retailers, and this data can be noisy as well - translations were done automatically and may contain inaccuracies.

Some SKUs already have a GTIN code available, which can be helpful for the final grouping or stand as a labelled data for ML-based approach.

The expected output should include two columns:

- SKU_ID – the unique identifier for each product
- ITEM_ID – the identifier representing the grouped product (i.e., all SKUs that refer to the same item should share the same ITEM_ID)

Columns Descriptions:

sku_id	product_id in specific webstore (in country X retailer combination)
retailer	webstore name
country	country of webstore
sku_name_eng	scraped product name in online retailer – translated into English
description_short_eng	product description (truncated to fixed maximum length) – translated into English
breadcrumbs_eng	native product categorization on retailer website – translated into English

sku_url	URL used to scrape all SKU info from retailer (may no longer be available)
image_url	image_URL pointing to the product scraped (may no longer be available)
sku_last_price	last scraped product price in Local Currency
gtin	*extracted/predicted* product GTIN code
brand	*extracted/predicted* product brand name (via ML model)
category	*extracted/predicted* product category (via ML model)
attribute	*extracted/predicted* product attributes (via ML models) in JSON format (**Attribute_1**: attribute_value_1, attribute_value_2; **Attribute_2**: attribute_value_3 ...)

In general, it is not required to use all columns in the dataset—feel free to choose the most relevant features for your approach.

Evaluation Criteria

There are multiple valid approaches to solving this task. Therefore, the most important aspect is to clearly explain the reasoning behind your chosen method—while still covering all other key components of the solution. Your submission will be evaluated based on the following criteria:

- Exploratory Data Analysis
- Explanation and reasoning behind the chosen solution logic
- Feature engineering (if applies)
- ML development (if applies)
- Code quality and structure
- Presentation skills
- Final results

The final output (a table with two columns: SKU_ID and ITEM_ID) will be compared to our internal product grouping. The **Rand Index** (match rate between your groupings and ours) will be estimated and shared.

The solution itself should be delivered in a notebook format.